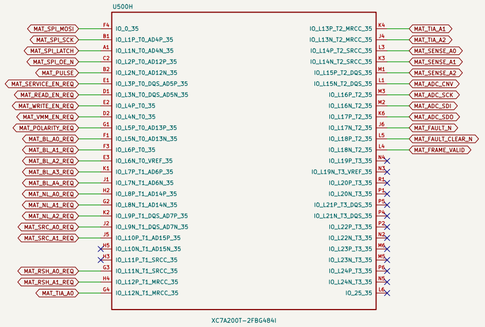
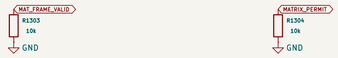
Matrix commands and hardware interlocks

FPGA MATRIX ENGINE / BANK 35

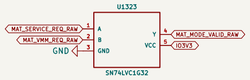


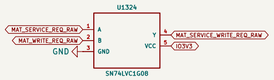


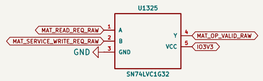
Fail-safe external permit / FPGA frame validator

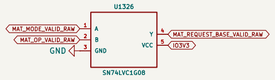


STRICT COMMAND VALIDATION



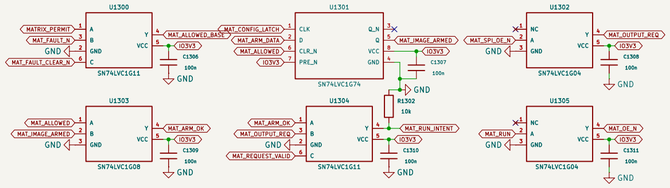




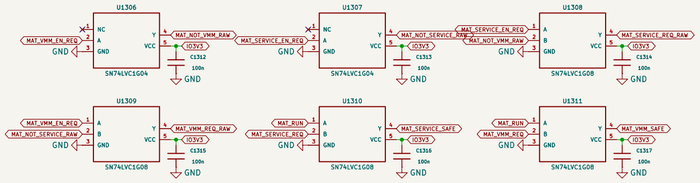




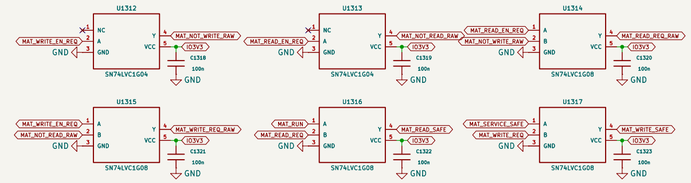
PERMIT / FRESH LATCH / OUTPUT ENABLE



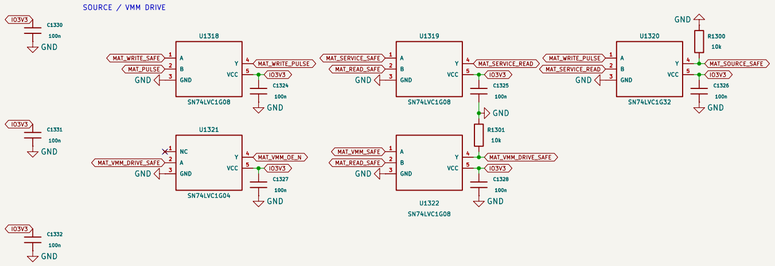
VMM / SERVICE MODE INTERLOCK



READ / WRITE REQUEST INTERLOCK



SOURCE / VMM DRIVE



Source qualification / write-DAC guard / fault event



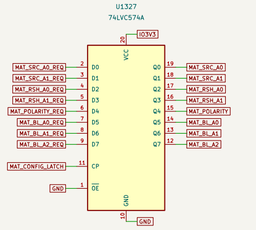




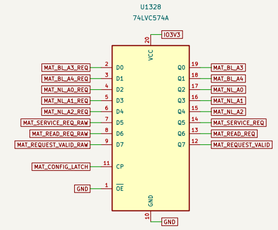


Serial Image: 1400
Analogue routing: 1600
AFE / ADC / fault: 1900–2200

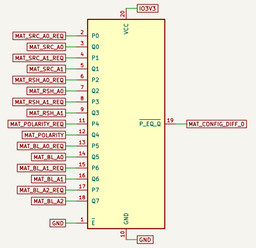
Atomic route capture / live-request comparison

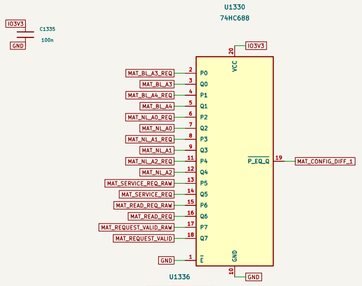














Power permission and safe defaults

MAT_ALLOWED requires external MATRIX_PERMIT, released fault clear and no latched fault. Power loss or clear resets frame arm. Capture registers have no defined startup state; their outputs cannot energize the matrix before a fresh qualified commit.

One atomic configuration commit

MAT_CONFIG_LATCH = MAT_SPI_LATCH & MAT_SPI_OE_N. Only a commit with outputs disabled clocks the 80-bit mask chain, both 8-bit configuration registers and frame arm. Keep LATCH low during OE changes. Hold all inputs stable around the qualified edge.

Routes held during operation

Two 74LVC574 registers capture SRC, RSH, POLARITY, BL and NL addresses plus exclusive service/read mode and request validity. Analog route controls use captured outputs. Two 74HC688 comparators detect request changes: mismatch blocks RUN and generates a latched fault request while output is intended.

Matched operation and source

SERVICE_READ requires source code 00. SERVICE_WRITE requires 01 SET, 10 RESET or 11 FORM. VMM_READ does not use the service-source code. SRC and RSH are held by U1327. All four shunts are available in service READ and WRITE. Change RSH only with source OFF and the route discharged.

Write command protection

CS_CTRL, CS_HV, DAC_LDAC_N and DAC_RESET_N must remain high throughout an intended WRITE pulse. A change suppresses the applied window and raises MAT_CONFIG_TRIP. VMM DAC chip selects are not included. READ may intentionally sweep its source DAC while the route remains fixed.

FPGA and serial-frame boundary

MAT_FRAME_VALID remains an HDL assertion: there is no external 80-clock counter, CRC or per-line mask checker. Validate exact frame length, no VMM/ground overlap, legal selected-line clamps and suitable WL selection. Address freeze does not validate the crossbar image.

Commit timing and re-arm

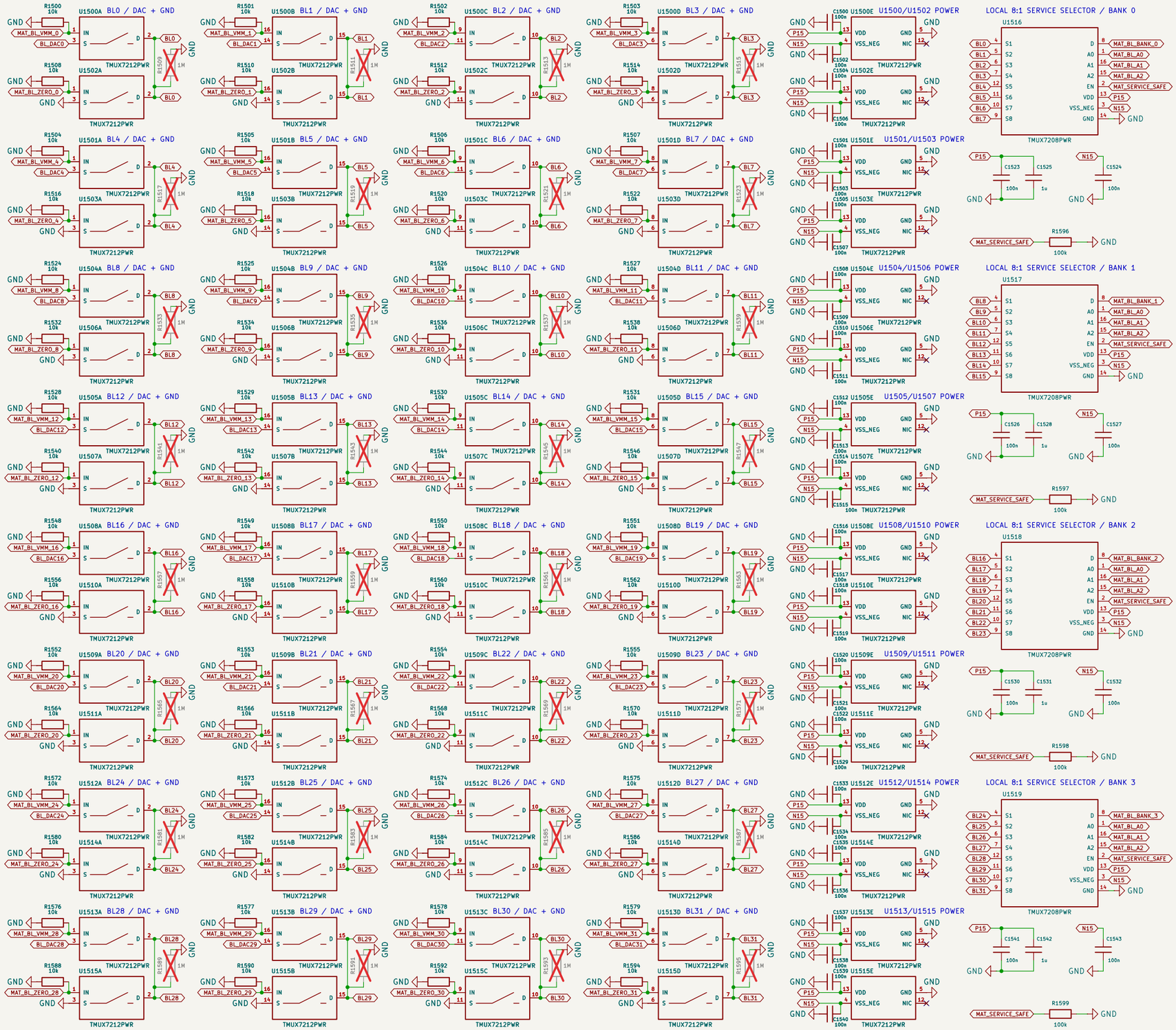
Use at least 1 us between the qualified LATCH edge and OE_N falling for the digital paths here; analog settling is additional and range dependent. A fault requires source zero, explicit clear release and a new full frame plus qualified commit. Never change OE while LATCH is high.

Stored energy and pulse timing

MAT_SOURCE_SAFE is SERVICE_READ or qualified WRITE_PULSE. It now controls the source actuator, not a charged pulse switch after current sense. Matrix capacitance and switch injection remain in the energy budget. FPGA bounds pulse duration and enforces zero-source discharge before rerouting.

MIRA
Sheet: /MATRIX/1300_MATRIX_FPGA_CONTROL_AND_INTERLOCKS/
File: 1300_MATRIX_FPGA_CONTROL_AND_INTERLOCKS.kicad_sch
Title: Matrix commands and hardware interlocks
Size: A2 | Date: 2026-10-18 | Rev: B
KiCad E.D.A. 10.0.8 | Id: 19/43

Bit-line DAC connections, clamps and service selection



Thirty-two bit-line terminals

This sheet contains BL0–BL31 in four groups of eight. Each terminal has two TMUX7212 paths: buffered DAC drive and GND clamp. VMM is voltage drive; there is no per-cell current compliance. Firmware must enforce the current and TIA limits from docs/pcb-preparation/analog-requirements/.

Normal drive and grounding

For drive, enable the DAC path and release its ground clamp; for GND, do the opposite. All 32 optional 1M BL bleeders are DNP to avoid unmeasured WRITE return current. With both switches OFF, a BL may float. Use BL_ZERO for a defined level or deliberate discharge. The 10k resistors bias control pins only.

Eight-to-one local selectors

U1516–U1519 each select one bit line from their eight-channel group. MAT_BL_A0, MAT_BL_A1 and MAT_BL_A2 are shared address inputs. The selector outputs are MAT_BL_BANK_0–3; the upper address selection on the service-router sheet chooses which bank reaches MAT_BL_SERVICE.

Service mode

MAT_SERVICE_SAFE enables the local service selectors. The service connection is separate from the VMM DAC switches, so firmware must prepare the bit-line masks before making a read, set, reset or forming connection. The chosen line must not be clamped against the active source.

Analog supply bypassing

The switch packages use P15 and N15 with local supply bypassing. BL6 and BL7 now use their dedicated BL_DAC6 and BL_DAC7 sources. All other DAC, clamp and service-selector connections are unchanged.

MIRA

Sheet: /MATRIX/1500_MATRIX_BL0_31_SWITCHING/
File: 1500_MATRIX_BL0_31_SWITCHING.kicad_sch

Title: Bit-line DAC connections, clamps and service selection

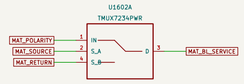
Size: A3	Date: 2026-10-18	Rev: B
KiCad E.D.A. 10.0.8		Id: 21/43

1600 / SERVICE COMMAND, ROUTING AND CONTROLLER RESET

VREAD / VSET / VRESET / VFORM COMMAND
SRC=00 READ, 01 SET, 10 RESET, 11 FORM.
U1601 no longer supplies DUT power.

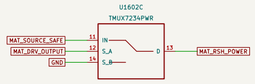


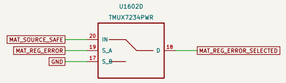
DUT POLARITY
1: BL=SOURCE, NL=RETURN
0: BL=RETURN, NL=SOURCE





UPSTREAM POWER ENABLE
OFF: shunt input to GND, ON: CV/CC driver.
No pulse switch after the current shunt.



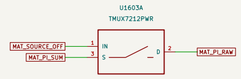


ERROR SELECTION
OFF selects zero; ON selects regulator error.
The PI controller and shunts are on sheet 2100.



RETURN PATH
READ: TIA virtual ground.
WRITE: direct GND; old low-side sink removed.





BL BANK SELECTION
Only the held service address is used.
VMM isolates the service path.



OFF RESET (MAT_SOURCE_OFF = NOT MAT_SOURCE_SAFE)
A clears the PI output-to-sum path.
D discharges the feedback capacitor directly.
R1605 limits direct capacitor-reset current.













SUPPLIES / LOCAL BYPASS
TMUX72xx use P15/N15. Respect rail sequencing; see MIRA.05 limits.









DNP BLEEDERS / PASSIVE CONTROL PULL-DOWNS
R1600/R1601 must remain DNP for microamp operation. Power OFF discharges SOURCE through the selected shunt.
Before route/range changes: source OFF; run an explicit clamp/discharge frame.
Source OFF isolates DUT lines; it does not prove their capacitance is discharged.









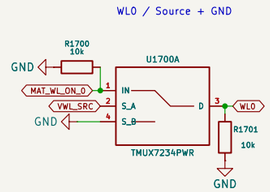


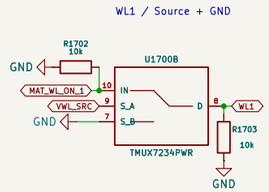
MIRA
Sheet: /MATRIX/1600_MATRIX_SERVICE_SOURCE_BANK_AND_POLARITY_ROUTING/
File: 1600_MATRIX_SERVICE_SOURCE_BANK_AND_POLARITY_ROUTING.kicad_sch

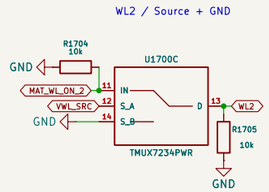
Title: Service source, bit-line bank and polarity routing

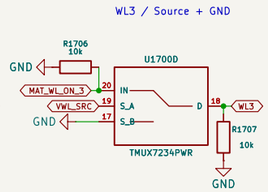
Size: A2	Date: 2026-10-18	Rev: B
KiCad E.D.A. 10.0.8		Id: 22/43

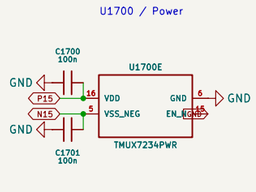
Word-line drive and neuron-line selection

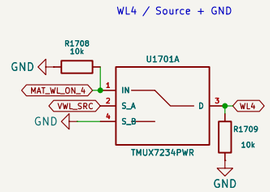


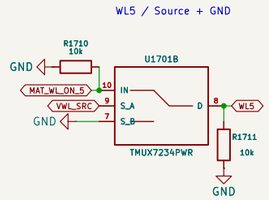


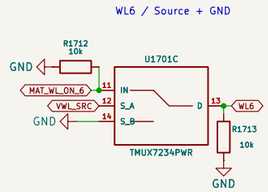


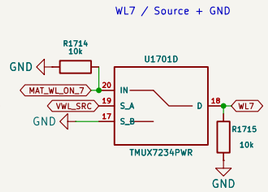


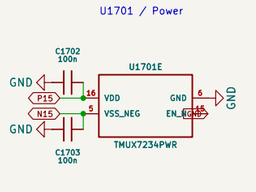


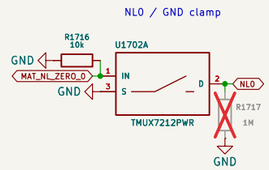


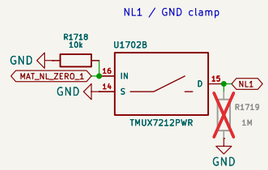


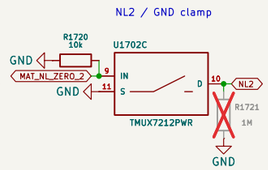


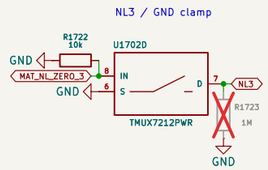


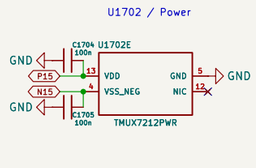


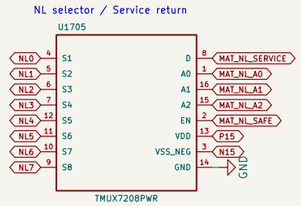


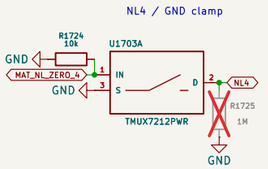


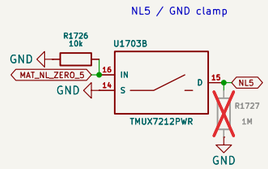


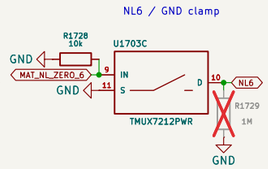


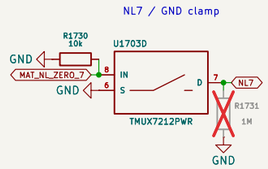


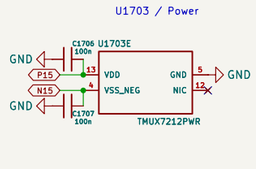


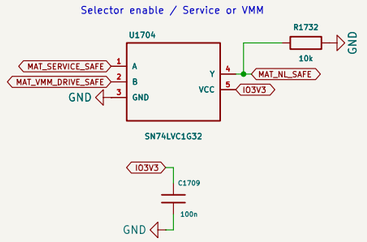












NL mask: U1409 on 1400
MAT_NL_SERVICE: polarity router 1600
One selected NL feeds the common TIA.
For VMM read, set MAT_POLARITY=1.
The selected NL clamp must be OFF.

WL operation
VWL_SRC: service DAC 1100
WL mask: U1408 on 1400
WL0–WL7: socket 1800

IN=0 selects GND.
IN=1 selects VWL_SRC.
Each terminal keeps its 10k load.

Eight word-line selectors

U1700 and U1701 are quad TMUX7234 switches. Each WL selects VWL_SRC or GND under MAT_WL_ON_0–7. VWL is a gate voltage only for a compatible 1T1R adapter. Validate VGS, VGD and gate oxide limits for BOTH polarities before programming; the socket pin map does not specify the DUT ratings.

Word-line loading

Each WL terminal retains its 10k load to GND. The control inputs also have 10k bias resistors. VWL_SRC comes from the service DAC sheet, so changing that DAC command changes the common available word-line drive level rather than selecting an individual row.

Independent neuron-line clamps

U1702/U1703 provide one ground-clamp switch per NL, controlled by MAT_NL_ZERO_0–7. All eight optional 1M NL bleeders are DNP: a fitted resistor would bypass WRITE current sensing. Release the selected clamp for measurement; use NL_ZERO deliberately for a defined level or discharge.

Shared measurement selector

U1705 is an eight-to-one selector. MAT_NL_A0–2 choose one neuron line, and its common output is MAT_NL_SERVICE. U1704 generates MAT_NL_SAFE by ORing MAT_SERVICE_SAFE with MAT_VMM_DRIVE_SAFE; either qualified operating path can therefore enable the selector.

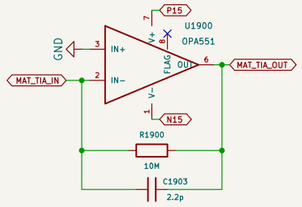
VMM return route

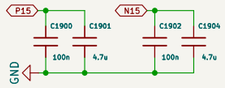
The selected neuron line continues through the polarity router and then to the common TIA during a read. The existing operating note specifies MAT_POLARITY=1 for VMM read. Only one NL is selected by this multiplexer at a time; the other neuron-line clamps remain independently programmable.

MIRA
Sheet: /MATRIX/1700_MATRIX_WL0_7_DRIVERS_AND_NL0_7_CLAMPS_SELECTOR/
File: 1700_MATRIX_WL0_7_DRIVERS_AND_NL0_7_CLAMPS_SELECTOR.kicad_sch
Title: Word-line drive and neuron-line selection
Size: A3 | Date: 2026-10-18 | Rev: B
KiCad E.D.A. 10.0.8 | Id: 23/43

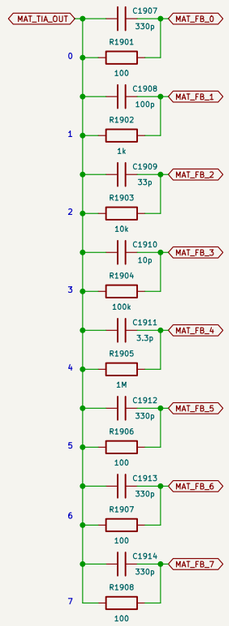
Matrix current conversion and monitor selection

TIA and permanent feedback

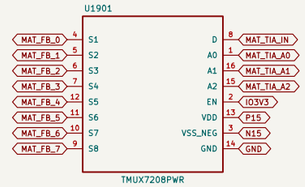


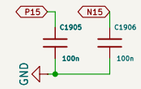


Paired Rf / Cf branches

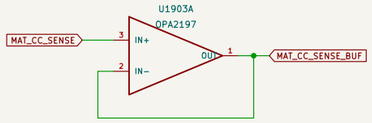


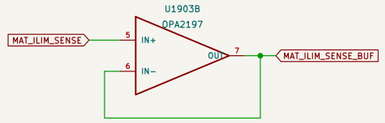
Feedback selector



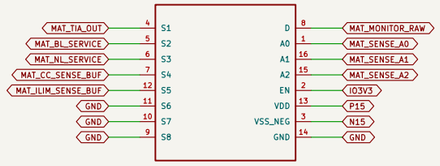


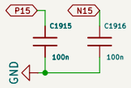
High−impedance taps for protection and monitor

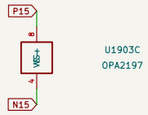




Voltage monitor selector







TIA gain and monitor address are independent



U1903: P15 / N15. Channel B also feeds the source regulator.

MAT_TIA_IN from service routing
MAT_MONITOR_RAW to ADC driver

Current−to−voltage conversion

U1900 holds MAT_TIA_IN near GND and converts input current into MAT_TIA_OUT. In the ideal low−frequency model VOUT = −IIN x REFF. REFF includes R1900 (10 MOhm) in parallel with the selected branch, including switch resistance. C1903 (2.2 pF) remains in feedback at every gain.

Selectable feedback branches

U1901 connects one paired resistor/capacitor branch across the amplifier. Address codes 0−4 select 100 Ohm/330 pF, 1 kOhm/100 pF, 10 kOhm/33 pF, 100 kOhm/10 pF, or 1 MOhm/3.3 pF. Codes 5−7 repeat the 100 Ohm/330 pF branch. Each selected branch operates in parallel with the permanent feedback network; switch resistance also affects the effective transimpedance.

Independent monitor address

U1902 selects TIA, BL service, NL service, buffered current sense or buffered ILIM. Codes 5−7 select GND. U1903A isolates the INA821 output; U1903B buffers the target for the controller, trip divider and monitor mux. The sheet 2000 buffer isolates the ADC gain network.

Measurement sequence and supplies

MAT_MONITOR_RAW feeds the OPA197 buffer on sheet 2000.
Initial validation: <=100 kSPS, >=100 us ADC blanking;
confirm settling after each monitor/TIA range change.
CV/CC needs separate ms−scale settling.
Reject abs(TIA_OUT)>=10 V; ADC may be below full scale.
Change TIA gain only with excitation OFF. P15/N15 supply.

MIRA

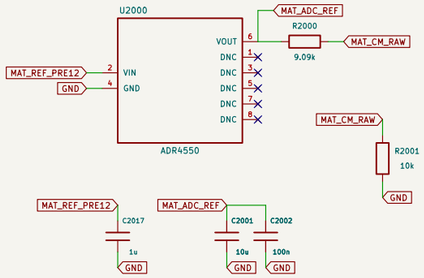
Sheet: /ADC/1900_ADC_TIA_AND_MONITOR_SELECTOR/
File: 1900_ADC_TIA_AND_MONITOR_SELECTOR.kicad_sch

Title: Matrix current conversion and monitor selection

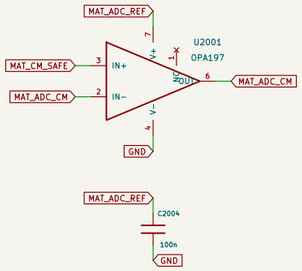
Size: A3	Date: 2026−10−18	**Rev: B**
KiCad E.D.A. 10.0.8		Id: 26/43

Buffered monitor and differential matrix ADC

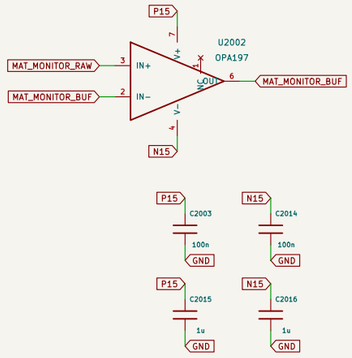
5 V reference from a protected 12 V input



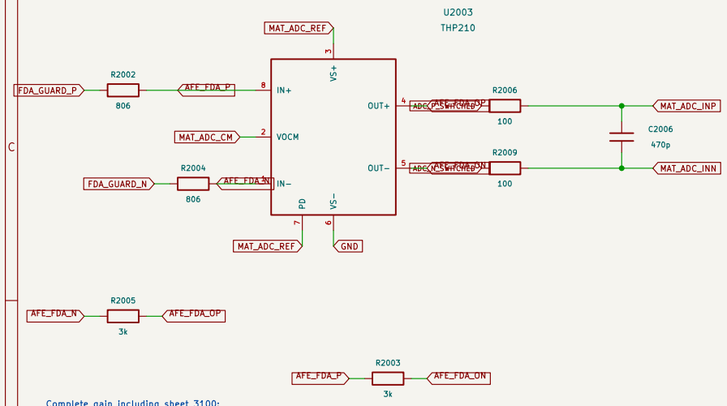
VREF / 2 common-mode buffer



High-Z post-mux buffer: P15 / N15

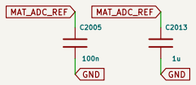


Balanced FDA gain and ADC filter

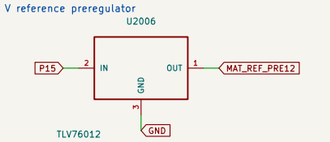


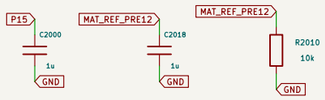
Complete gain including sheet 3100:
VDIFF / VMON = −0.148935704 nominal.
C2006 is between IN+ and IN−.

Matched FDA resistor ratios: see sheet 3100.
PD and VS+ share MAT_ADC_REF.



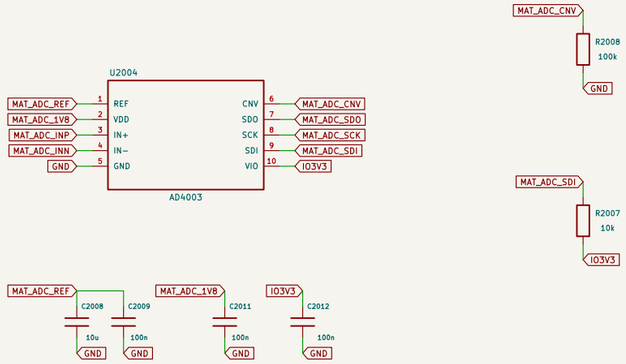
12 V reference preregulator



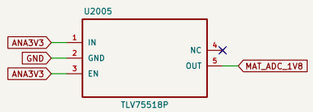


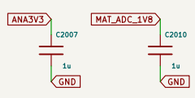
TLV76012 pin 1 OUT / pin 2 IN / pin 3 GND
R2010 guarantees a load above 1 mA.

ADC and serial interface



Local ADC 1.8 V supply





Reference input headroom

U2006 derives a nominal 12 V input from P15 for U2000. This keeps the ADR4550 input away from its 15 V upper operating boundary. R2010 provides the preregulator minimum load; the 5 V reference is not powered from ANA5.

High-impedance monitor input

U2002 follows MAT_MONITOR_RAW on P15/N15 before the resistive FDA network. Its high-impedance input isolates the selected node from the FDA gain network. Raw CC and ILIM are separately buffered by U1903 on sheet 1900; S4/S5 observe those buffered nodes.

True differential conversion

U2003 and U2001 share MAT_ADC_REF with the ADC reference. Sheet 3100 adds balanced attenuation and power-off isolation. Nominal VDIFF = −0.148935704 × VMON, average about 2.5 V. At VMON = +/−15.75 V, outputs are 1.327...3.673 V and FDA input common mode is 1.591...2.920 V. Qualify settling and reference loading on the prototype; see v6 report.

Filter and validation timing

R2006/R2009 are symmetric 100 Ohm isolation resistors; C2006 is 470 pF C0G between ADC inputs. Differential RC tau is 94 ns. Begin validation at <=100 kSPS with 100 us minimum blanking after monitor/TIA-range changes. Confirm actual settling and loop stability; these initial settings are not a verified 2 MSPS or CV/CC settling specification.

Serial and local power

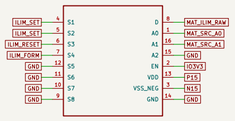
AD4003 uses MAT_ADC_1V8 for VDD and IO3V3 for its interface. CNV, SCK, SDI and SDO retain their U500 connections. R2008 holds CNV low; R2007 holds SDI high. Use gain −0.148935704 and per-range calibration. REF readiness clears the matrix image.

MIRA
Sheet: /ADC/2000_ADC_CONVERTER_AND_PRECISION_REFERENCES/
File: 2000_ADC_CONVERTER_AND_PRECISION_REFERENCES.kicad_sch
Title: Buffered monitor and differential matrix ADC
Size: A3 | Date: 2026-10-18 | **Rev: B**
KiCad E.D.A. 10.0.8 | Id: 27/43

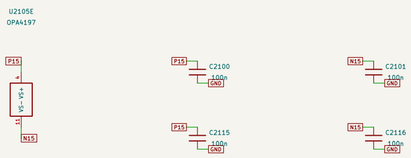
Bias is approximately ANA5/2 (loaded divider); command ceiling = 2.01 x MAT_ADC_REF (nominal 10.05 V). All analog amplifiers use P15/N15.

Scope: source-side CV/CC includes DUT, leakage and charging. Microsecond pulse accuracy and transient peak current are NOT qualified; see MIRA.05.

Current command selection



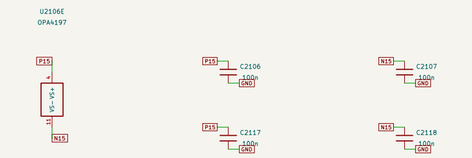
00 READ / 01 SET / 10 RESET / 11 FORM
READ and SET share ILIM_SET. Reload only with outputs OFF.



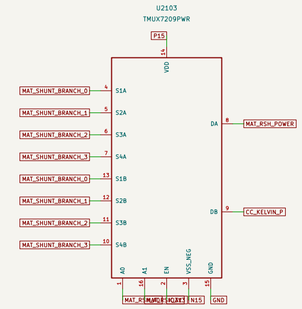
Four source shunts



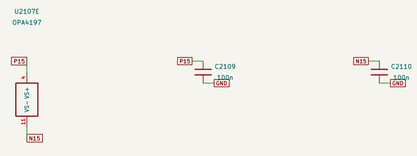
RSH: 100 / 1k / 10k / 100k: all four in service READ/WRITE.
Initial profile uses 0.08…0.8 V sense; standard current 10 uA…8 mA.
Nominal full-DAC capability is about 10 mA: calibrate zero/gain.



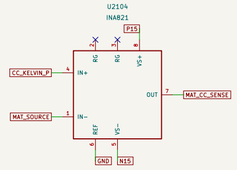
Kelvin range selector



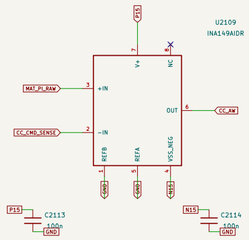
A = force path; B = high-Z Kelvin sense
EN stays high; switch ranges only with source OFF.



Current sense / high-CM anti-windup

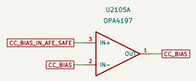


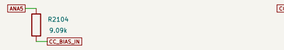




Bias reference

BIAS is shared by both error cells and cancelled before the PI.

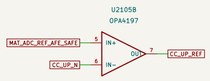


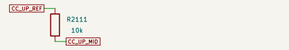




Common-mode ceiling

10.05 V ceiling uses the ADC precision reference through a high-Z input.





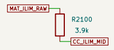


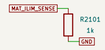


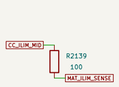
Command sense buffer

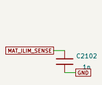
CBUF isolates CMD from clamp feedback resistor loads.





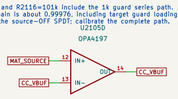


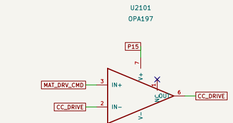


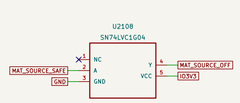


Voltage sense and source driver

R2115=100k and R2116=101k include the 1k guard series path.
Nominal CV gain is about 0.99976, including target guard loading.
Driver feeds the source-OFF SPDT; calibrate the complete path.







Voltage error

VERR = BIAS + VSOURCE − VTARGET.





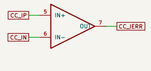






Current error

IERR = BIAS + ISENSE − buffered ILIM target.











Precision maximum: voltage

D2100 provides main and local anti-saturation feedback.





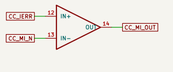


Precision maximum: current

D2101 selects the larger shifted error without a floating branch.







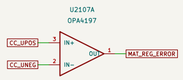






Remove error bias

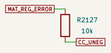
ERROR = max(VERR, IERR) − BIAS.











Shared PI and back-calculation

Sheet1600 resets PI_SUM to RAW and to C in source-OFF state.

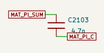








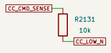


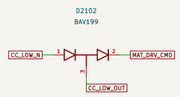


Command lower clamp

CBUF feedback keeps the clamp branch linear while inactive.





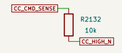


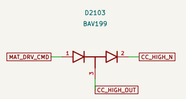
Lower limit 0 V; upper command nominal 10.05 V. CBUF isolates clamp feedback. INA149 back-calculation measures RAW − buffered CMD.

Command upper clamp

Ceiling = 2.01 x MAT_ADC_REF; actual output depends on load drop.







MIRA

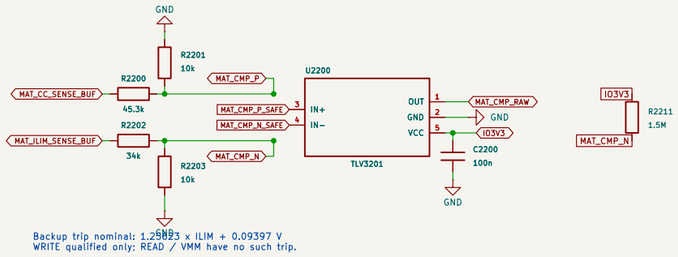
Sheet: /PROTECTION/2100_PROTECTION_WRITE_CURRENT_COMPLIANCE/
File: 2100_PROTECTION_WRITE_CURRENT_COMPLIANCE.kicad_sch

Title: Source-side CV/CC, Kelvin current sense and anti-windup

Size: A2	Date: 2026-10-18	Rev: B
KiCad E.D.A. 10.0.8		Id: 29/43

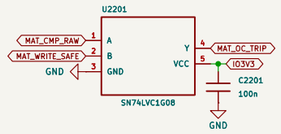
Write overcurrent trip, fault memory and power inhibit

Sense dividers and comparator

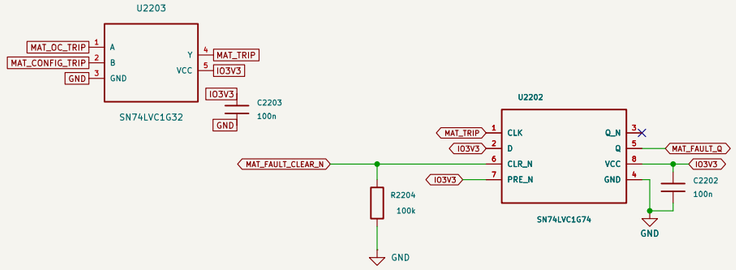


Backup trip nominal: 1.25823 x ILIM + 0.09397 V
WRITE qualified only; READ / VMM have no such trip.

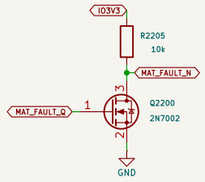
Qualified trip event



Clocked trip memory / asynchronous clear

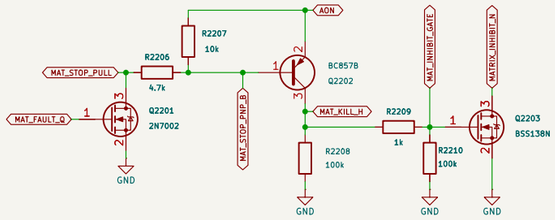


FPGA fault output



MAT_FAULT_N to FPGA Interlock

Power−module inhibit driver



MATRIX_INHIBIT_N to Power / connector sheet 400

Protected overcurrent scale

R2200/R2201 = 45.3k/10k; R2202/R2203 = 34k/10k. Include both 100k guard bleeds on sheet 3000 and R2211=1.5M from IO3V3. Nominal backup trip: VSENSE > 1.25623*VILIM + 0.09397 V at IO3V3=3.3 V. This is coarse shutdown, not precision CC. Comparator offset, leakage and resistor tolerances add error.

Input headroom and fault current

At +16.5 V sense, IN+ is at most about 2.76 V, inside the valid common−mode range for IO3V3 >= 3.135 V. Negative saturation or a powered−off comparator is current limited by the input resistors; calculated clamp current is below 0.5 mA versus the 10 mA absolute limit. Normal source sense and limit are 0..1 V. Do not use the fault comparator as a calibrated low−current meter.

Analog and configuration trips

U2201 keeps analog qualification: MAT_OC_TRIP = MAT_CMP_RAW & MAT_WRITE_SAFE. U2203 ORs this event with MAT_CONFIG_TRIP from sheet 1300. Route/range/mode changes and service−DAC updates during intended WRITE now clock the same fault memory as excess current. The configuration branch is not masked by WRITE_SAFE.

Fault memory and re−arm

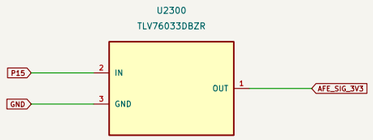
U2202 stores D=1 on a MAT_TRIP rising edge. PRE_N stays high; only CLR_N is asynchronous. R2204 defaults clear low, also disabling MAT_ALLOWED on sheet 1300. Clear release alone cannot arm the matrix; source zero, a full validated 80−bit frame and protected LATCH are required. Meet latch clock−pulse and clear recovery/removal timing.

Local stop and Power inhibit

MAT_FAULT_Q pulls MAT_FAULT_N low through Q2200 and drives Q2201/Q2202/Q2203 to assert MATRIX_INHIBIT_N to Power. Local enable removal does not wait for MCU or ADC. Total shutdown delay includes sense amplifier, input RC, comparator, gates, latch, source actuator and switch timing; the 40 ns comparator headline is not the system stopping time.

MIRA
Sheet: /PROTECTION/2200_PROTECTION_MATRIX_TRIP_AND_FAULT_INTERFACE/
File: 2200_PROTECTION_MATRIX_TRIP_AND_FAULT_INTERFACE.kicad_sch
Title: Write overcurrent trip, fault memory and power inhibit
Size: A3 | Date: 2026−10−18 | **Rev: B**
KiCad E.D.A. 10.0.8 | Id: 30/43

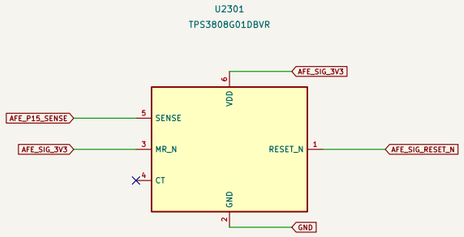
2300_ANALOG_SIGNAL_QUALIFIER



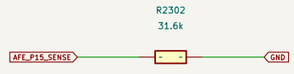




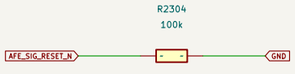




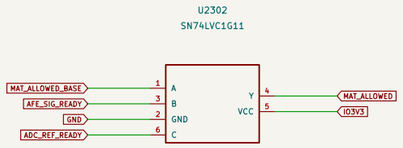




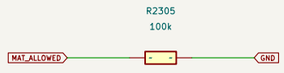




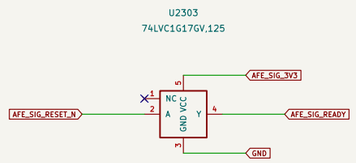


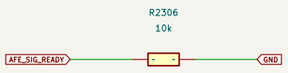








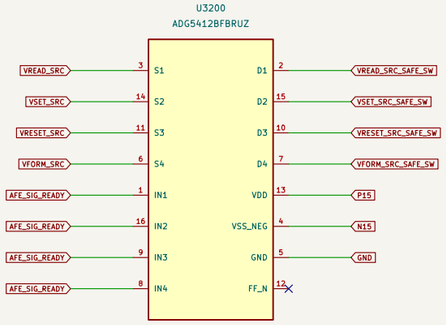


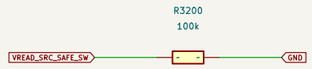


Local P15 threshold ~13.22 V; R2301/R2302 are 0.1%; CT open gives 20 ms nominal release delay.
Loss of AFE_SIG_READY clears the Control image latch; recovery requires fresh configuration.
Guard switching precedes image arming. This is independent of PM software.

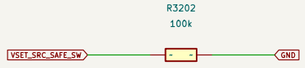
Sheet: /2300_ANALOG_SIGNAL_QUALIFIER/		
File: 2300_ANALOG_SIGNAL_QUALIFIER.kicad_sch		
Title:		
Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 33/43

3200_FORMING_SOURCE_ISOLATION

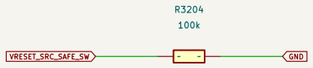




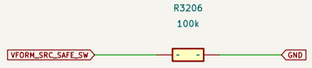


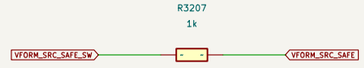
















All four service targets use the same guarded path. BF protects both S and D.
The guard changes state only at local power qualification, before matrix image arming.
R2115=100k / R2116=101k include the 1k series guard path. Calibrate total CV gain.
This prevents retained PROG rails feeding an unpowered P15/N15 source bank.

Sheet: /3200_FORMING_SOURCE_ISOLATION/
File: 3200_FORMING_SOURCE_ISOLATION.kicad_sch

Title:

Size: A3 | Date: | **Rev:**
KiCad E.D.A. 10.0.8 | Id: 42/43